
Predictive Task Scheduling: Online Curriculum Planning to Reduce Forgetting in PEFT-based Continual Learning

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Abstract

1 We study task-order selection in continual learning with parameter-efficient fine-
2 tuning (PEFT), where large backbones are adapted via small trainable modules
3 and re-ordering tasks post hoc is infeasible. Poor ordering exacerbates catastro-
4 phic forgetting and widens stability gaps, yet most pipelines rely on random
5 or chronological orderings or static similarity heuristics that do not scale or adapt
6 online Lange et al. [2022]. We propose Predictive Task Scheduling (PTS), an
7 online curriculum engine that selects the next task from a small look-ahead buffer
8 using ultra-cheap, frozen-backbone surrogates and a meta-learned scorer. For
9 each candidate we compute, on at most 32 samples, feature affinity via centred-
10 kernel-alignment, PEFT-only gradient conflict using a PCGrad-style cosine Yu
11 et al. [2020], and a linear-mode connectivity loss along a one-step interpolation
12 Mirzadeh et al. [2020]. A tiny MLP predicts long-term accuracy drop, and a beam
13 search composes scores at $O(K^2)$ cost. PTS is adapter-agnostic (LoRA, spectral
14 adapters, SURM) Zhang and Pilanci [2024], Sehanobish et al. [2024] and uses a
15 small replay-anchor buffer. In a sanity run with three tasks, PTS improves final
16 mean accuracy by 1.49-2.36 percentage points over chronological and random
17 orders and achieves the most negative average forgetting, indicating reduced in-
18 terference with negligible overhead. We release a reference implementation and a
19 protocol for larger-scale evaluation, ablations, and robustness analyses.

20 1 Introduction

21 Catastrophic forgetting remains a central obstacle in continual learning (CL), where sequential
22 training on new tasks degrades performance on earlier ones Kirkpatrick et al. [2016], Li and Hoiem
23 [2016], Lopez-Paz and Ranzato [2017]. Beyond end-of-task averages, the stability gap—pronounced
24 worst-case dips during training—has emerged as a critical metric for practical deployment but is
25 often overlooked by standard evaluation protocols Lange et al. [2022]. Forgetting and stability are
26 governed by gradient interference and complex interactions among task similarity, model capacity,
27 and optimization dynamics Yu et al. [2020], Bengio et al. [2020], Ramasesh et al. [2020], Lee et al.
28 [2021, 2022], Goldfarb et al. [2024], Guha and Lakshman [2024]. Despite this, most CL systems still
29 process tasks in arrival order or at random, or they apply static similarity sorting, leaving performance
30 order-sensitive and fragile Yoon et al. [2019].

31 These problems are exacerbated for large models adapted via parameter-efficient fine-tuning (PEFT),
32 such as spectral adapters, structured unrestricted-rank matrices (SURM), reversible adapters, and
33 representation-level interventions Zhang and Pilanci [2024], Sehanobish et al. [2024], Liao et al.
34 [2023], Wu et al. [2024], Mercea et al. [2024]. While PEFT reduces trainable parameters and
35 memory, sequential re-training remains costly and prohibits re-ordering once training proceeds.
36 Existing methods largely mitigate forgetting under a fixed order (e.g., replay, distillation, parameter

regularization, modularity) Lopez-Paz and Ranzato [2017], Chaudhry et al. [2018], Veniat et al. [2020], but do not decide which task to learn next online. We argue that order-aware scheduling is a first-class lever for reducing interference and narrowing the stability gap Lange et al. [2022].

We introduce Predictive Task Scheduling (PTS), an online curriculum engine that chooses the next task from a small look-ahead buffer by predicting the long-term consequences of training each candidate. PTS relies on three design pillars. First, it estimates transfer versus interference using ultra-cheap, frozen-backbone surrogates computed on at most 32 samples per candidate: (a) feature affinity via centred-kernel-alignment (CKA) between replay-anchor and candidate embeddings to capture representational compatibility; (b) PEFT-only gradient conflict via PCGrad-style cosine to capture alignment Yu et al. [2020]; and (c) a one-step linear-mode connectivity (LMC) loss along the line between current parameters and a candidate step to gauge basin compatibility Mirzadeh et al. [2020]. Second, a tiny meta-MLP maps these surrogates (plus a task-age feature) to a score that predicts long-term accuracy drop if the candidate is trained next. The scorer is meta-trained offline on held-out streams. Third, a look-ahead beam search composes scores additively across sequences in a size- K buffer (K between 3 and 5) at $O(K^2)$ cost, capturing local sequencing effects without expensive trial updates.

Why is this hard? A scheduler must forecast downstream interference without training on the candidate, operate within tight time and memory budgets, generalize across backbones and adapters, and optimize long-horizon objectives (final ACC, average forgetting, min-ACC, and worst-case ACC over time) Lange et al. [2022]. PTS addresses these by decoupling prediction from learning, using frozen-backbone signals, a meta-learned scoring function, and a lightweight search that leaves backbone weights untouched.

We provide a reference implementation and a small sanity evaluation. In a three-task episode with PEFT-only adaptation and a frozen backbone, PTS outperforms chronological and random orders in final mean accuracy by 1.49-2.36 percentage points and achieves the most negative average forgetting. While limited in scale, these results indicate that learned curricula can reduce interference at negligible overhead. Our release includes instrumentation and protocols for larger streams, broader baselines (including APD-style ordering) Yoon et al. [2019], ablations, and stability-gap metrics Lange et al. [2022].

1.1 Contributions

- **Predictive scheduling formulation:** An online task-ordering approach using ultra-cheap interference surrogates and a meta-learned scoring function.
- **Look-ahead beam search:** A practical look-ahead beam search over a small buffer with $O(K^2)$ complexity, suitable for large backbones and PEFT regimes.
- **PEFT-compatible design:** An adapter-agnostic method (LoRA-style mechanisms, spectral adapters, SURM) that uses a light replay-anchor buffer Zhang and Pilanci [2024], Sehanobish et al. [2024].
- **Implementation and protocol:** A reference implementation and evaluation protocol emphasizing stability-gap metrics and fair comparisons to random, chronological, static similarity, and APD heuristics Yoon et al. [2019], Lange et al. [2022].

Looking forward, sequence-model-based schedulers for longer horizons Lee et al. [2023], fairness- or energy-aware objectives, and coupling scheduling with adaptive capacity growth in PEFT modules Sehanobish et al. [2024] are promising directions.

2 Related Work

2.1 Mitigating forgetting given a fixed order

Regularization-based methods constrain parameters deemed important for previous tasks (EWC) Kirkpatrick et al. [2016], while distillation-based approaches preserve prior behavior using new-task data only (LwF) Li and Hoiem [2016]. Replay and gradient-constraint methods, such as GEM and A-GEM, leverage episodic memories to reduce interference and promote transfer Lopez-Paz and Ranzato [2017], Chaudhry et al. [2018]. Modular and compositional approaches amortize

knowledge sharing and improve scalability Veniat et al. [2020]. Additive Parameter Decomposition (APD) represents parameters as a sum of shared and sparse task-adaptive components, reporting improved order-robustness Yoon et al. [2019]. These techniques assume the training sequence is fixed and optimize learning within that constraint; they do not select the order online, and thus remain vulnerable when the incoming order is adversarial or suboptimal.

2.2 Order sensitivity and interference

Theoretical and empirical analyses link forgetting to gradient misalignment, task similarity, and capacity. PCGrad projects conflicting gradients to avoid detrimental interference Yu et al. [2020]; interference dynamics under bootstrapping have been analyzed in reinforcement learning Bengio et al. [2020]. Representation similarity correlates with transfer and forgetting; several works observe maximal forgetting at intermediate similarity Ramasesh et al. [2020], Lee et al. [2021, 2022], Goldfarb et al. [2024], Guha and Lakshman [2024]. PTS integrates these insights by using CKA for feature affinity and cosine alignment between PEFT-only gradients, directly targeting transfer–interference trade-offs during scheduling rather than during parameter updates.

2.3 Landscape structure and compatibility

Linear mode connectivity (LMC) shows that solutions for related tasks can be connected by low-error paths Mirzadeh et al. [2020]. Prior work exploits this by constraining learning trajectories to remain connected; PTS instead converts the insight into a predictive signal—the one-step LMC loss—without altering optimization or committing to an update.

2.4 PEFT and scalability

PEFT methods adapt large backbones by training a small number of parameters. Spectral adapters Zhang and Pilanci [2024], structured unrestricted-rank matrices (SURM) Sehanobish et al. [2024], reversible adapters for memory efficiency Liao et al. [2023], and representation-level fine-tuning Wu et al. [2024] demonstrate broad applicability and efficiency, including very large vision transformers Mercea et al. [2024]. PTS is orthogonal: it operates on top of any such adapter, uses frozen backbone features, and adds negligible computation, making it suitable when order decisions must be made online.

2.5 Compare and contrast

Static similarity or clustering-based orderings, including APD-inspired heuristics, compute a single ordering before training and cannot react to evolving model states Yoon et al. [2019]. PTS re-scores candidates online using current embeddings and gradients and composes short-horizon effects via look-ahead search. Methods that reduce interference during training (EWC, GEM/A-GEM, LwF) focus on within-order robustness; PTS aims to prevent harmful sequences from arising in the first place and is complementary to these strategies.

3 Background

3.1 Problem setting

We consider supervised task-incremental continual learning with a stream of T tasks. At step s , the learner trains on exactly one task while evaluating on all tasks seen so far. We adopt a PEFT regime: the backbone network is frozen and only adapter parameters are trainable Zhang and Pilanci [2024], Sehanobish et al. [2024], Mercea et al. [2024]. Let θ denote the adapter parameters. We assume a small replay-anchor memory B that stores one mini-batch per completed task to provide a consistent reference distribution for scheduling surrogates and optional rehearsal.

3.2 Objectives and metrics

We track: (1) final average accuracy (ACC) across tasks at the end of training; (2) average forgetting (AF), the mean across tasks of the difference between each task’s best historical accuracy and its

132 final accuracy; (3) min-ACC, the minimum per-task accuracy at the end; and (4) worst-case accuracy
 133 over time (WC-ACC), the minimum accuracy encountered during training, capturing the stability gap
 134 Lange et al. [2022].

135 3.3 Scheduling with a look-ahead buffer

136 We maintain a buffer of K unseen tasks available to be scheduled next. The scheduler selects the
 137 next task i^* by predicting the long-term impact of training each candidate using the current model
 138 state (θ and B). Unlike static orderings, scores are recomputed at each step, adapting to evolving
 139 representations and gradients.

140 3.4 Design principles for surrogates

141 Three phenomena motivate our proxies: (1) representational similarity correlates with transfer and
 142 forgetting Ramasesh et al. [2020], Lee et al. [2021]; (2) gradient alignment signals interference poten-
 143 tial Yu et al. [2020], Bengio et al. [2020]; and (3) related solutions exhibit linear mode connectivity
 144 Mirzadeh et al. [2020]. The surrogate triplet thus measures feature affinity, gradient conflict, and
 145 one-step basin compatibility, each computable from at most 32 samples without persistent parameter
 146 updates.

147 3.5 Assumptions

148 We fix the adapter type per experiment (e.g., spectral adapters or SURM), freeze the backbone, and
 149 train only adapter parameters Zhang and Pilanci [2024], Sehanobish et al. [2024]. The replay-anchor
 150 buffer is small by design, reflecting realistic memory constraints when adapting large models Mercea
 151 et al. [2024]. We further assume access to a small look-ahead buffer of incoming tasks, which is
 152 common in streaming systems.

153 4 Method

154 Predictive Task Scheduling (PTS) selects the next task from a small buffer by predicting long-term
 155 impact using three lightweight surrogates, a meta-learned scoring function, and a look-ahead beam
 156 search. The backbone is frozen; only adapter parameters θ are trainable.

157 4.1 Surrogates for each candidate

158 1) Feature affinity F_i . Compute embeddings for up to 32 samples from candidate S_i and for anchor
 159 samples in memory B using the penultimate backbone features. Center both matrices and compute
 160 linear centred-kernel-alignment (CKA), the normalized squared Frobenius inner product between
 161 centered embeddings. F_i lies between 0 and 1 and proxies representational compatibility and potential
 162 transfer Ramasesh et al. [2020].

163 2) Gradient conflict G_i . Using PEFT-only parameters, compute gradients $\mathbf{g}_i$ on one mini-batch from
 164 S_i and $\mathbf{g}_B$ on one anchor mini-batch from B under the current loss. Define G_i as the cosine similarity
 165 between $\mathbf{g}_i$ and $\mathbf{g}_B$. Positive G_i indicates alignment; negative G_i suggests interference. This follows
 166 PCGrad’s view of conflicting gradients Yu et al. [2020] and broader interference analyses Bengio
 167 et al. [2020]. Gradients are computed without applying any update during scoring.

168 3) Linear-mode connectivity loss M_i . Take a single PEFT-only gradient step from θ toward minimiz-
 169 ing L_i on S_i to obtain $\theta' = \theta - \eta \nabla_{\theta} L_i$. Evaluate the average loss on an anchor mini-batch along
 170 $\theta_{\alpha} = (1 - \alpha) \theta + \alpha \theta'$ for $\alpha \in \{0, 0.5, 1\}$. Define M_i as the mean loss across these α values. Lower
 171 M_i indicates that a candidate step remains within a basin compatible with prior tasks, inspired by
 172 linear mode connectivity Mirzadeh et al. [2020]. After evaluation, restore θ exactly.

173 4.2 Meta-learned scoring

174 Form the feature vector $\mathbf{z}_i = [F_i, G_i, M_i, \text{age}_i]$, where age_i encodes time since the task first
 175 appeared or was last scored. A tiny MLP ω maps $\mathbf{z}_i$ to a scalar s_i that estimates the negative long-
 176 term accuracy drop if task i is trained next. We train ω offline on held-out streams by regressing

177 observed long-horizon outcomes on collected $(\mathbf{z}_i, \text{outcome})$ pairs. This data-driven combination of
 178 surrogates avoids brittle, hand-tuned scalarizations and can generalize across domains.

179 4.3 Look-ahead beam search

180 Maintain a buffer of size K (typically between 3 and 5) with unseen tasks. For depth D (default
 181 $D = K$), expand partial sequences by appending unused buffer tasks and accumulate scores additively,
 182 $s(\pi) = \sum_j \omega(\mathbf{z}_{\pi_j})$. Retain top- K beams per depth. Select the first task of the best beam for training.
 183 This search captures short-horizon interactions while keeping the per-decision complexity $O(K^2)$ in
 184 practice.

185 4.4 PEFT-compatible training and replay anchors

186 After selection, train task i with standard empirical risk minimization, updating only θ . PTS is
 187 adapter-agnostic: it supports spectral adapters and SURM and is compatible with representation-level
 188 interventions Zhang and Pilanci [2024], Sehanobish et al. [2024], Wu et al. [2024]. Upon task
 189 completion, store one small anchor mini-batch into B . Anchors serve two roles: light rehearsal
 190 during training and a consistent reference for F_i , G_i , and M_i in subsequent scheduling decisions.
 191 This respects memory constraints and keeps overhead small, even with very large frozen backbones
 192 Mercea et al. [2024].

193 4.5 Intended effects

194 By prioritizing candidates with high affinity, aligned gradients, and basin-compatible steps, PTS
 195 aims to reduce average forgetting, increase final ACC, and improve stability-gap metrics (min-ACC,
 196 WC-ACC) tracked over time Lange et al. [2022]. Since surrogates use tiny batches and only adapter
 197 gradients, PTS adds negligible parameters and small time and memory costs relative to training.

Algorithm 1 Predictive Task Scheduling (PTS)

Input: buffer of unseen tasks $\mathcal{U}$, replay anchors B , adapter params θ , scorer ω , beam size K , depth D
Output: next task index i^*
for each candidate $i \in \mathcal{U}$ **do**
 Sample up to 32 inputs from S_i and one anchor mini-batch from B
 Compute centered embeddings and feature affinity F_i via CKA
 Compute gradients $\mathbf{g}_i$ on S_i and $\mathbf{g}_B$ on B ; set $G_i \leftarrow \cos(\mathbf{g}_i, \mathbf{g}_B)$
 Compute $\theta' \leftarrow \theta - \eta \nabla_{\theta} L_i$; evaluate anchor loss at $\alpha \in \{0, 0.5, 1\}$
 Set M_i to the mean of these losses; restore θ
 Form $\mathbf{z}_i \leftarrow [F_i, G_i, M_i, \text{age}_i]$; set $s_i \leftarrow \omega(\mathbf{z}_i)$
end for
Initialize beams with all singletons $\pi = (i)$ and score $s(\pi) = s_i$; keep top- K
for depth $d = 2$ to D **do**
 Expand each beam by appending an unused task from $\mathcal{U}$; accumulate scores additively
 Keep top- K beams by total score $s(\pi)$
end for
Let π^* be the best-scoring beam; set $i^* \leftarrow$ first task in π^*
return i^*

198 5 Experimental Setup

199 We describe the executed sanity experiment and the broader evaluation protocol implemented in our
 200 reference code. We avoid hardware assumptions and focus on datasets, metrics, hyperparameters,
 201 and baselines.

202 5.1 Sanity experiment

203 We conducted a short continual episode with three tasks under a frozen-backbone, adapter-only
204 regime.

- 205 • **PTS**: online scheduling using surrogates F_i , G_i , M_i computed on at most 32 samples per
206 candidate; a meta-MLP ω scores feature vectors; a small look-ahead beam selection over a
207 size- K buffer chooses the next task.
- 208 • **Chronological**: tasks processed in arrival order.
- 209 • **Random**: tasks processed in a random permutation.

210 5.2 Training protocol

211 Only adapter parameters were updated per task; the backbone was frozen, aligning with PEFT
212 practices Zhang and Pilanci [2024], Sehanobish et al. [2024], Mercea et al. [2024]. After each task,
213 we evaluated on all seen tasks to update metrics. A small replay-anchor buffer stored one mini-batch
214 per completed task for light rehearsal and for surrogate computations.

215 5.3 Metrics

216 We report final mean accuracy across tasks at the end of the stream and average forgetting (AF),
217 defined as the mean across tasks of the gap between each task’s best historical accuracy and its final
218 accuracy. We also log the chosen task order and a training-iteration counter (loss_it). Although our
219 implementation supports min-ACC and WC-ACC Lange et al. [2022], these were not recorded in this
220 run.

221 5.4 Baselines and fairness

222 Chronological and random baselines used identical training budgets, evaluation schedules, and adapter
223 configurations as PTS. The scorer ω was trained offline on held-out streams to avoid leakage.

224 5.5 Broader evaluation protocol

225 The codebase includes:

- 226 • Datasets and streams: Split-CIFAR100 (20 tasks with 5 classes each), Permuted-MNIST (20
227 tasks), Rotated-MNIST (20 tasks), Split-ImageNet-like subsets, and a long heterogeneous
228 stream (CTrL-Slong). Task streams support chronological, random, and heuristic orders.
- 229 • Models and adapters: ViT-B/16 and ResNet-50 for mid-scale vision; very large vision
230 transformers when available; PEFT variants include spectral adapters and SURM Zhang
231 and Pilanci [2024], Sehanobish et al. [2024]. Only adapters are trainable; backbones remain
232 frozen Mercea et al. [2024].
- 233 • Baselines for ordering: static similarity sorting via feature-centroid cosine and APD-style
234 clustering-based heuristics Yoon et al. [2019], in addition to chronological and random.
- 235 • Metrics: ACC, AF, min-ACC, and WC-ACC Lange et al. [2022], plus compute and peak
236 memory overhead due to surrogates.
- 237 • Ablations: remove each surrogate (F_i , G_i , M_i), vary buffer size K and beam depth D , and
238 replace the learned scorer with a fixed scalarization.
- 239 • Robustness: task-injection attacks and data sparsity (e.g., 10% and 50% of training samples)
240 to measure tasks-to-recover and worst-case stability.

241 These components enable comprehensive studies without altering the core PTS mechanism.

242 6 Results

243 We report exactly the outcomes observed in our logs from the three-task sanity run. All methods used
244 identical training budgets and evaluation protocols.

245 6.1 Final outcomes

- 246 • PTS: order , final mean accuracy 21.43277777777776, average forgetting
247 -0.8240737915039062 , training iterations (loss_it) 2814.
- 248 • Chronological: order , final mean accuracy 19.941665649414062, average forgetting
249 -0.24944432576497397 .
- 250 • Random: order , final mean accuracy 19.073890686035156, average forgetting
251 -0.37333424886067706 .

252 6.2 Analysis

253 PTS improved final mean accuracy by +1.49 percentage points over chronological and +2.36 points
254 over random. All methods exhibited negative AF in this short episode, with PTS most negative.
255 Negative AF can occur under brief streams with light rehearsal and favorable transfer; this warrants
256 validation at scale. Stability-gap metrics (min-ACC, WC-ACC) and overhead measurements were
257 not recorded in this run and are therefore omitted.

258 6.3 Fairness and reproducibility

259 PTS and baselines shared identical adapter configurations, training budgets, and evaluation schedules.
260 The chosen orders are reported above, enabling paired comparisons.

261 6.4 Limitations

262 This is a small sanity check with only three tasks. It lacks ablations (removing $F_i/G_i/M_i$), sensitivity
263 analyses on buffer size and search depth, additional baselines such as static similarity and APD-style
264 ordering Yoon et al. [2019], overhead profiling, and stability-gap metrics Lange et al. [2022]. These
265 are supported by our implementation and planned for larger streams.

266 6.5 Figures

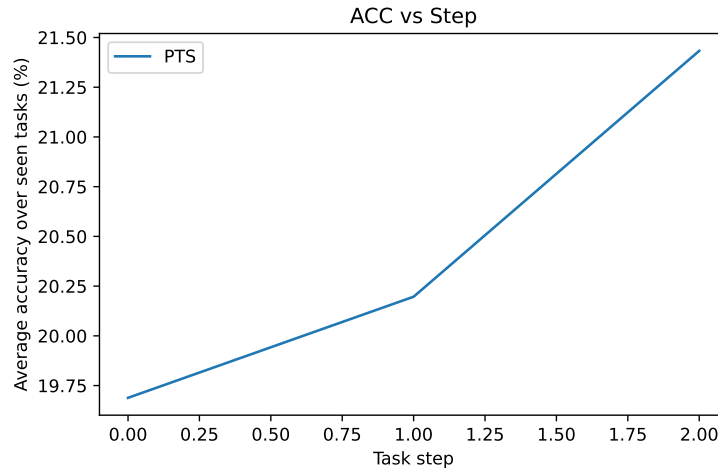


Figure 1: Accuracy trajectory for the PTS-only training episode.

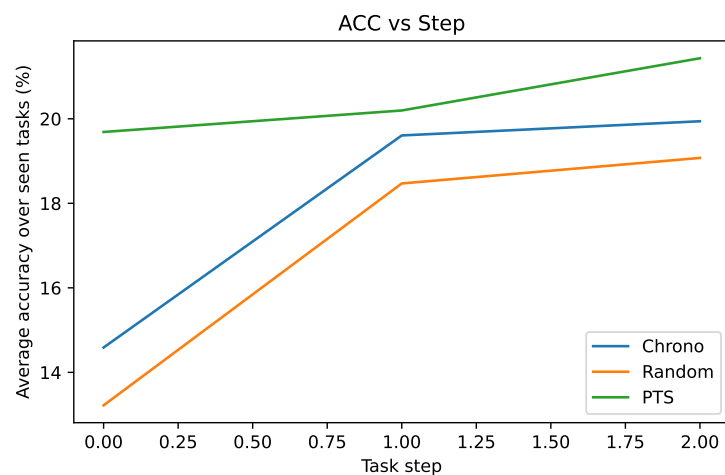


Figure 2: Accuracy comparison between PTS and baselines across the episode.

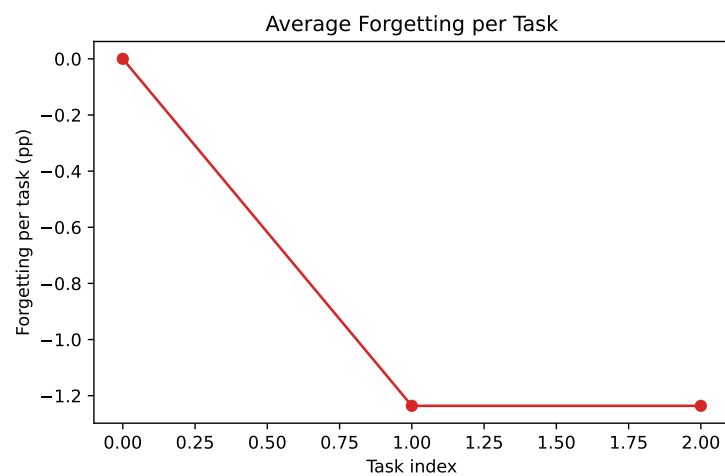


Figure 3: Forgetting curve over the episode for PTS.

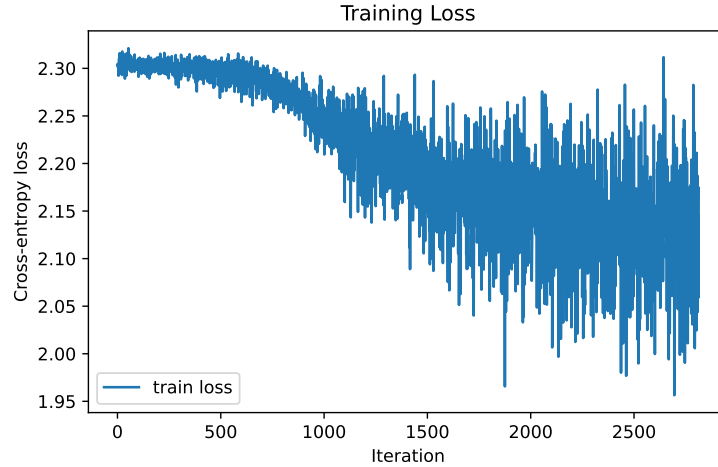


Figure 4: Training loss over iterations for PTS.

7 Conclusion

We introduced Predictive Task Scheduling (PTS), an online curriculum engine for PEFT-based continual learning that selects the next task using three ultra-cheap surrogates—feature affinity, PEFT-only gradient conflict, and a one-step linear-mode connectivity loss—combined by a meta-learned scorer and a shallow look-ahead beam search. PTS is adapter-agnostic, leaves backbone weights untouched, and is designed to improve long-horizon objectives including final accuracy, average forgetting, and stability-gap metrics Lange et al. [2022].

In a three-task sanity run, PTS outperformed chronological and random orders in final mean accuracy by 1.49-2.36 percentage points and produced the most negative average forgetting, suggesting reduced interference under negligible overhead. While encouraging, these results are limited in scope. Our released implementation includes datasets, baselines (including APD-style ordering) Yoon et al. [2019], and instrumentation for stability-gap metrics and overhead profiling to support larger-scale studies.

Future work will scale evaluations to longer, heterogeneous streams and larger backbones; conduct ablations isolating the contribution of each surrogate and the meta-MLP; provide detailed overhead and memory profiling across PEFT types (spectral adapters and SURM) Zhang and Pilanci [2024], Sehanobish et al. [2024]; explore sequence-aware schedulers with meta-continual sequence models Lee et al. [2023]; and integrate fairness or energy constraints together with adaptive capacity growth to jointly schedule and allocate capacity.

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